

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Data Visualization with Power BI and Tableau

Practical – IX

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Class: TYIT	Batch: 1
Date of Assignment: 01/09/2026	Date/Time of Submission: 01/09/2026

Cleaning and Structuring Data for Analysis

a) Identify and handle missing or inconsistent data

Physics_Clean ×

```
IFNULL(FLOAT(REPLACE(STR([Physics]), '%', '')), 0)
```

The calculation is valid.

Apply

OK

Attendance_Clean ×

```
FLOAT(TRIM(REPLACE(STR([Attendance]), '%', '')))
```

The calculation is valid.

Apply

OK

Math_Clean ×

```
IFNULL([Math], 0)
```

The calculation is valid.

Apply

OK

Chemistry_Clean ✕

IFNULL([Chemistry], 0)

▶

The calculation is valid.

Apply

OK

b) Convert wide data into tall format

#	#		
Sem1_Scores.csv	Sem1_Scores.csv		ng
Math	Physics		
85	0.900000		95
nuff	0.750000		85
78	0.820000		88
92	0.880000		96
45	nuff	50	60
88	0.910000	85	92

Abc	#
Pivot	Pivot
Subject	Score
Chemistry	88.0000
Math	85.0000
Physics	0.9000
Programming	95.0000
Chemistry	80.0000
Math	nuff

Sem1_Scores.csv 10 fields 40 r

Name

✓ Number (decimal)

Number (whole)

Date & Time

Date

String

Boolean

✓ Default

Geographic Role ▶

Describe

No des

Fields

Type

Abc

Abc

#

em...

tten...

ivot ...

ivot ...

c) Perform union of multiple files

Union

Specific (manual)

Wildcard (automatic)

Connection: Sem2_Scores

Sem1_Scores.csv

Sem2_Scores.csv

Tables in union: 2

Apply

OK

Abc

Union

Table Name

Sem1_Scores.csv

Sem2_Scores.c...

Sem2_Scores.c...

Sem2_Scores.c...

Sem2_Scores.c...

Sem2_Scores.c...

Semester

```

IF CONTAINS([Table Name], "Sem1") THEN "Semester 1"
ELSEIF CONTAINS([Table Name], "Sem2") THEN "Semester 2"
ELSE "Other"
END

```

The calculation is valid.

Apply

OK

d) Execute cross-database joins

Sem1_Scores

1 Alert



Sem1_Scores

Connection
☒ Live ☐ Extract
 Filters
 0 | [Add](#)



Union — Departme...

4 rows

Name

Department_Master.csv

How do relationships differ from joins? [Learn more](#)

Union

Operator

Department_Maste...

Abc Dept_Info

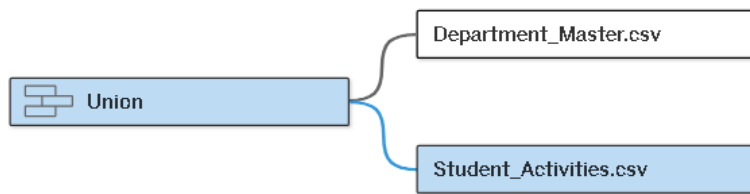
=

Abc Dept Code

Add more fields

Performance Options

Abc	Abc	Abc	#	#
Department_Master.csv	Department_Master.csv	Department_Master.csv	Department_Master.csv	Department_Master.csv
Dept Code	Department Name	HOD Name	Dept Target Average	Total Dept Capacity
CSE-01	Computer Science	Dr. Raman	85	120
ECE-02	Electronics & Comm	Dr. Iyer	78	90
IT-03	Information Tech	Dr. Sen	82	60
MECH-04	Mechanical Engg	Dr. Sharma	70	60



Union — Student_A...

10

Name Student_Activities.csv

How do relationships differ from joins? [Learn more](#)

Union

Operator

Student_Activities...

Abc Student_ID

=

Abc Student ID (Stui

⊕ Add more fields

> Performance Options

Remove Relationship

Search

🔍

🔼

🔽

Abc Student_Activities.csv

Abc Student_Activities.csv

Student_Activities.csv

Student_Activities.csv

Student ID (Student Acti	Club Membership	Projects Completed	Certifications
S101	Coding Club	3	2
S102	Robotics Club	2	1
S103	Coding Club	2	1
S104	Debate Society	4	3
S105	Sports Club	1	0
S106	Coding Club	3	2

Tables

- # Dept Target Average
- # Total Dept Capacity
- # Department_Master.csv (...)
- Abc Student_Activities.csv
 - Abc Club Membership
 - Abc Student ID (Student Act..
 - # Certifications
 - # Projects Completed
 - # Student_Activities.csv (C...
- Union
 - Abc Attendance
 - Abc Dept_Info
 - Abc Full_Name
 - Abc Semester
 - Abc Student_ID
 - Abc Table Name
 - # Attendance_Clean
 - # Chemistry
 - # Chemistry_Clean
 - # Math
 - # Math_Clean
 - # Physics
 - # Physics_Clean
 - # Programming
 - # Union (Count)
- Abc Measure Names

e) Restructure data for better visualization

Total_Score ×

`[Math_Clean] + [Physics_Clean] + [Chemistry_Clean] + [Program`

The calculation is valid. Apply OK

Overall_Percentage ×

`[Total_Score] / 400.0`

The calculation is valid. Apply OK

Performance_Band ×

`IF [Overall_Percentage] >= 0.85 THEN "Distinction"
ELSEIF [Overall_Percentage] >= 0.75 THEN "First Class"
ELSEIF [Overall_Percentage] >= 0.60 THEN "Second Class"
ELSE "Needs Academic Support"
END`

The calculation is valid. Apply OK

Attendance_Status ×

`IF [Attendance_Clean] >= 75 THEN "Eligible (Good Standing)"
ELSE "Shortage Warning (<75%)"
END`

The calculation is valid. Apply OK

Pages	Columns	Semester
	Rows	Student_ID Full_Name

Filters

Marks

Automatic

Color

Size

Text

Detail

Tooltip

Performance_Ba...

ATTR(Attendanc...

SUM(Overall_Per...

Sheet 1

Student..	Full_Name	Semeste..	Semeste..
S101	Arav Sharma	67.23%	69.23%
S102	Priya Verma		62.45%
	Priya VERMA	41.44%	
S103	Rohan Gupta	41.71%	63.21%
S104	Sneha Patel	69.72%	71.98%
S105	Aditya Rao	38.75%	43.15%
S106	Ananya Singh In	66.48%	68.49%
S107	Rahul Nair	38.93%	58.68%
S108	Kavya Reddy	71.49%	72.74%
S109	Vikram Joshi	49.40%	51.91%
S110	Pooja Iyer	62.96%	65.47%

Performance_Band

Needs Academic Supp..

Second Class

f) Handle different levels of detail in datasets

Dept_Average_Percentage

×

{ FIXED [Dept_Info] : AVG([Overall_Percentage]) }

▶

The calculation is valid.

Apply

OK

Default Number Format [Dept_Average_Percentage_LOD]

×

Automatic

Number (Standard)

Number (Custom)

Currency (Standard)

Currency (Custom)

Scientific

Percentage

Custom

Percentage

Decimal places:

1

▲

▼

Clear

OK

Cancel

Student_Overall_Career_Ave

{ FIXED [Student_ID] : AVG([Overall_Percentage]) }

The calculation is valid.

Apply

OK

Default Number Format [Student_Overall_Career_Average]

Automatic
Number (Standard)
Number (Custom)
Currency (Standard)
Currency (Custom)
Scientific
Percentage
Custom

Percentage
Decimal places:
1

Clear

OK

Cancel

Variance_From_Dept_Ave

[Overall_Percentage] - [Dept_Average_Percentage_LOD]

The calculation is valid.

Apply

OK

Default Number Format [Variance_From_Dept_Average]

Automatic
Number (Standard)
Number (Custom)
Currency (Standard)
Currency (Custom)
Scientific
Percentage
Custom

Percentage
Decimal places:
1

Clear

OK

Cancel

Benchmark_Status



```
IF [Variance_From_Dept_Average] >= 0 THEN "Above Department Benchmark"
ELSE "Below Department Benchmark"
END
```



The calculation is valid.

Apply

OK

Pages

Filters

Marks

- Automatic
- Color
- Size
- Text
- Detail
- Tooltip
- Benchmark_Status
- SUM(Dept_Average)
- SUM(Overall_Percentage)

Columns

- Semester

Rows

- Dept_Info
- Student_ID
- Full_Name

Sheet 2

Dept_Info	Student..	Full_Name	Semester	
			Semeste..	Semeste..
CSE-01	S101	Aarav Sharma	67.23%	69.23%
	S103	Rohan Gupta	41.71%	63.21%
	S106	Ananya Singh	66.48%	68.49%
	S109	Vikram Joshi	49.40%	51.91%
ECE-02	S102	Priya VERMA	41.44%	
		Priya Verma		62.45%
	S107	Rahul Nair	38.93%	58.68%
IT-03	S110	Pooja Iyer	62.96%	65.47%
	S104	Sneha Patel	69.72%	71.98%
MECH-04	S108	Kavya Reddy	71.49%	72.74%
	S105	Aditya Rao	38.75%	43.15%

Benchmark_Status

- Above Department Benchmark
- Below Department Benchmark

g) Apply advanced data cleaning techniques

Full_Name_Clean



```
TRIM([Full_Name])
```



The calculation is valid.

Apply

OK

First_Name

×

TRIM(SPLIT([Full_Name], " ", 1))

The calculation is valid.

Apply

OK

Last_Name

×

TRIM(SPLIT([Full_Name], " ", -1))

The calculation is valid.

Apply

OK

Dept_Branch

×

SPLIT([Dept_Info], "-", 1)

The calculation is valid.

Apply

OK

Section_ID

×

SPLIT([Dept_Info], "-", 2)

The calculation is valid.

Apply

OK

Student_Display_Label

×

[Student_ID] + " - " + [Full_Name_Clean] + " (" + [Dept_Branc



The calculation is valid.

Apply

OK

▼ **Student_Activities.csv**

- Abc Club Membership
- Abc Student ID (Student Act...
- # Certifications
- # Projects Completed
- # Student_Activities.csv (C...

▼ **Union**

- Abc Attendance
- =Abc Attendance_Status
- =Abc Benchmark_Status
- =Abc Dept_Branch
- Abc Dept_Info
- =Abc First_Name
- Abc Full_Name
- =Abc Full_Name_Clean
- =Abc Last_Name
- =Abc Performance_Band
- =Abc Section_ID
- =Abc Semester
- =Abc Student_Display_Label
- Abc Student_ID
- Abc Table Name
- =# Attendance_Clean
- # Chemistry
- =# Chemistry_Clean

Pages

Filters

Marks

iii Columns

Rows

Section_ID

Automatic

Color

Size

Text

Detail

Tooltip

Performance_Ba...

SUM(Overall_...

Sheet 3

	Semester	
Student_Display_La...	Semeste..	Semeste..
S101 - Aarav Sharm..	67.23%	69.23%
S102 - Priya Verma (..		62.45%
S102 - Priya VERMA ..	41.44%	
S103 - Rohan Gupta ..	41.71%	63.21%
S104 - Sneha Patel (I..	69.72%	71.98%
S105 - Aditya Rao (..	38.75%	43.15%
S106 - Ananya Singh..	66.48%	68.49%
S107 - Rahul Nair (E..	38.93%	58.68%
S108 - Kavya Reddy ..	71.49%	72.74%
S109 - Vikram Joshi ..	49.40%	51.91%
S110 - Pooja Iyer (EC..	62.96%	65.47%

Performance_Band

Needs Academic Supp..

Second Class